UF Herbert Wertheim College of Engineering
UNIVERSITY of FLORIDA

Lecture 4B – Physical Ergonomics: Work Physiology II

EIN 3241: Human Factors and Ergonomics
Spring 2026

February 9, 2026

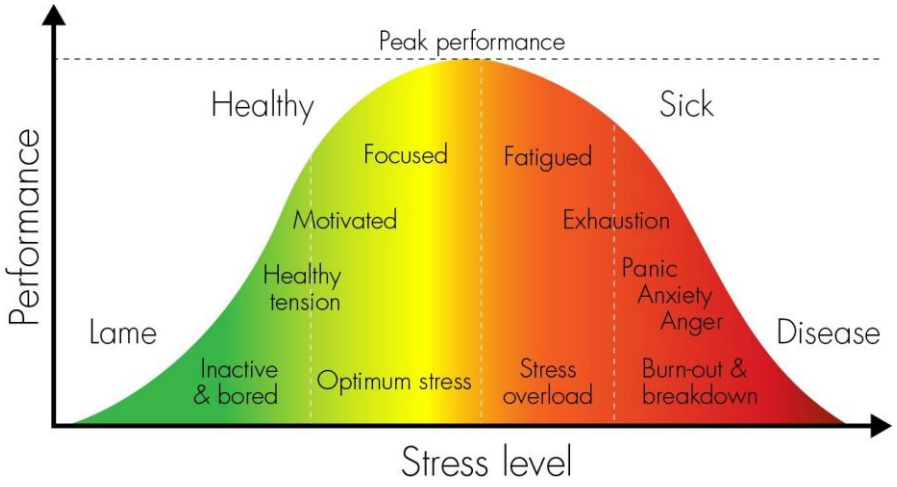
Dr. Jad A. Atweh

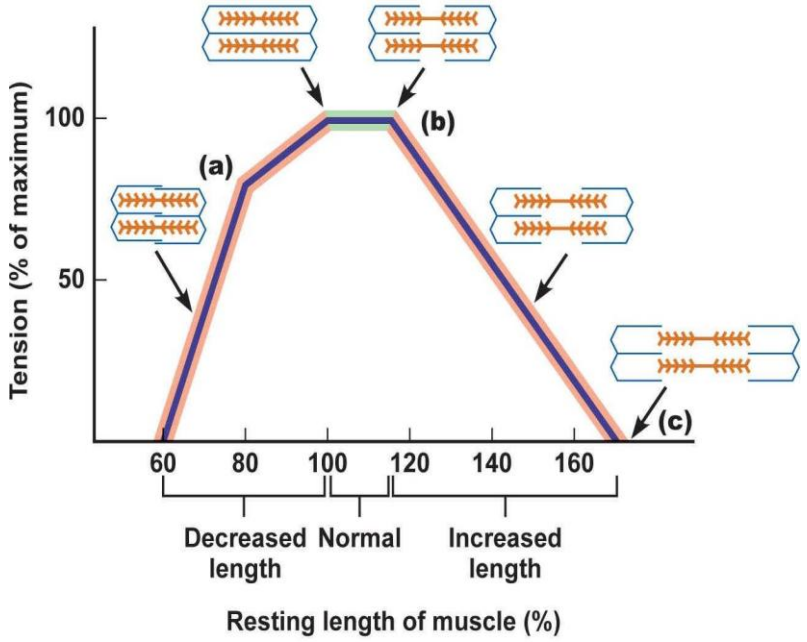
Admin Issues

- Lab 1 due Wednesday
- Homework 3 and Solution posted!
- Preparation for Occupational Biomechanics: Brush up on statics
  - https://www.youtube.com/watch?v=OffvP-cpMnY "Simplification of Forces and Moments"
  - https://www.youtube.com/watch?v=NrL5d-2CabQ "Vector Addition of Coplanar Forces (x-y components)"
  - https://www.youtube.com/watch?v=F8IHrg3pc7g "How to Solve a Scalar Triple Product" (2D mainly)
  - https://www.youtube.com/watch?v=QNNnPZ68STI "Moment of a Force"
  - A lot more…
    - https://www.youtube.com/watch?v=rdK0c9YZRYw "Moment About a Specified Axis" (2D mainly)

Recap

- Stress & Strain
  - Physical
  - Psychomotor
  - Cognitive
- Stress & Performance
- Bones
- Muscles
- Length-Force Relationship
- Muscular Effort
- EMG…
- Metabolism
- Supporting Systems
- ….





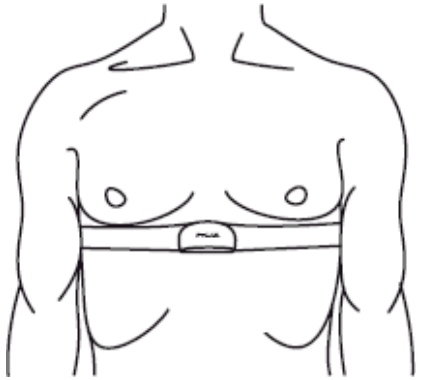
Energy Cost of Work

- What measures implicate an increase in physical work?
  - Energy demands increase
  - Oxygen consumption increases
  - Heart rate increases
  - Breathing rate accelerates
  - Tidal volume increases
  - Blood pressure increases
  - Increased heat production causing sweating to dissipate heat

- Stress due to a physical activity can be evaluated using one or preferably a combination of the above objective measures

Energy Cost of Work: O_2 Consumption and Heart Rate

- Oxygen consumption (VO_2) and heart rate are the indices most often used to measure physical strain on the body.



Energy Cost of Work: O_2 Consumption and Heart Rate

- Resting heart rate (HR) averages between 60 and 85 BPM.
- With increasing physical work, HR rises to meet O_2 demands.

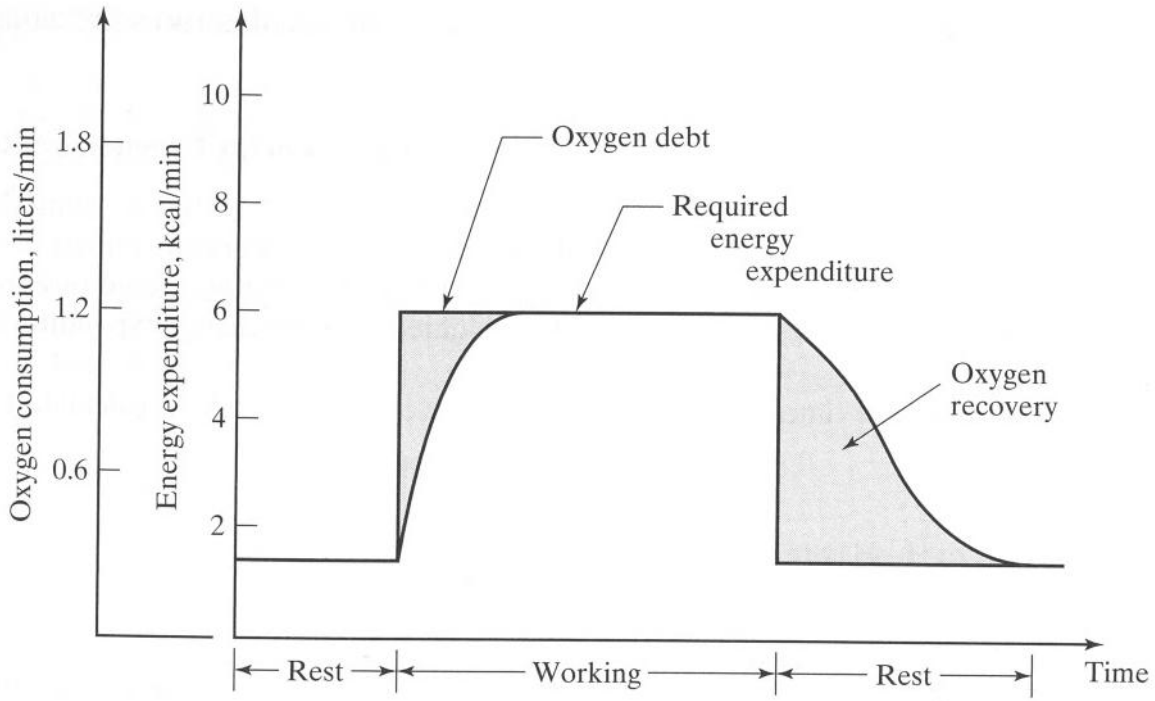


Figure 23.3 Oxygen debt accumulated at the beginning of work activity, which must be repaid when the activity stops.

Energy Cost of Work: O_2 Consumption and Heart Rate

- Suppose T1 and T2 are two different tasks performed by the same person, which task is more difficult?
- Suppose T1 and T2 are two different people performing the same task, which person is in better physical shape?

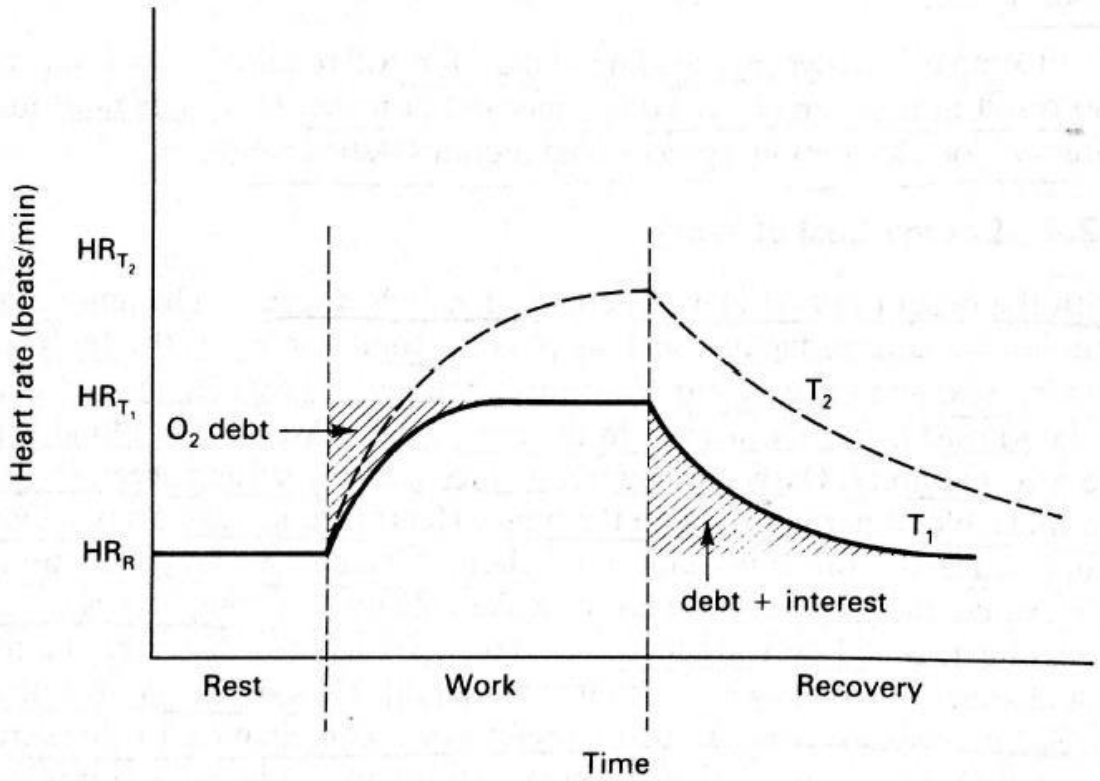


Figure 3.5 Heart rate with two tasks under submaximal work.

Energy Cost of Work: O_2 Consumption and Heart Rate

- There is a maximum attainable heart rate for any person, the primary determinant is age.
- At the age of 65, one can attain only about 60% of the heart rate at the age 20.
- Estimated Max HR = 220 – Age (years)
- $VO_2\ max$ prediction based on resting HR:

$$VO_2\ max = 15.3\ \times \frac{HR_{max}}{HR_{resting}}$$

- $HR_{reserve}(bpm) = HR_{max} - HR_{resting}$

- $\%\ of\ Reserve\ HR(bpm) = \frac{HR_{activity} - HR_{resting}}{HR_{max} - HR_{resting}} \times 100$

Energy Cost of Work: O_2 Consumption and Heart Rate

- Every liter of oxygen consumed, there is an average of 4.8kcal of energy expended
- The linear relationship between O_2 consumption and HR has been demonstrated in laboratory and field studies:

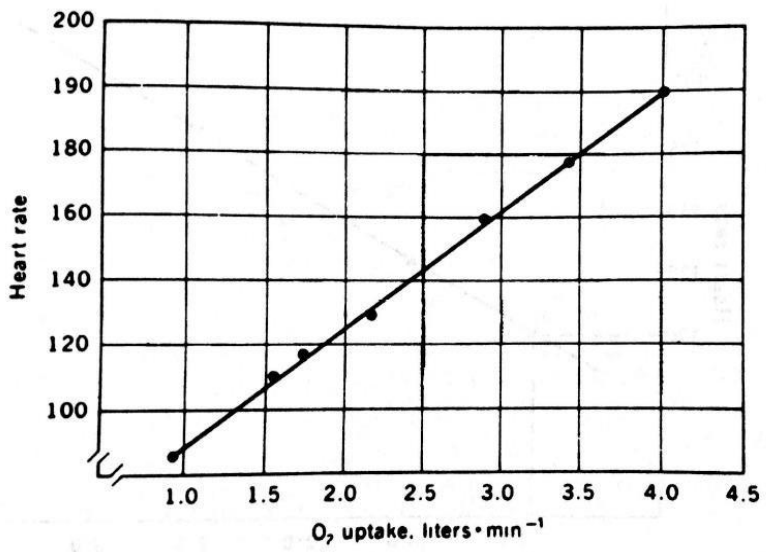


Figure 3.6 The relationship between heart rate and O_2 consumption on the cycle ergometer. (From *Textbook of Work Physiology, 3/E*, P.O. Astrand and K. Rodahl, 1986, used with permission of McGraw-Hill, Inc.)

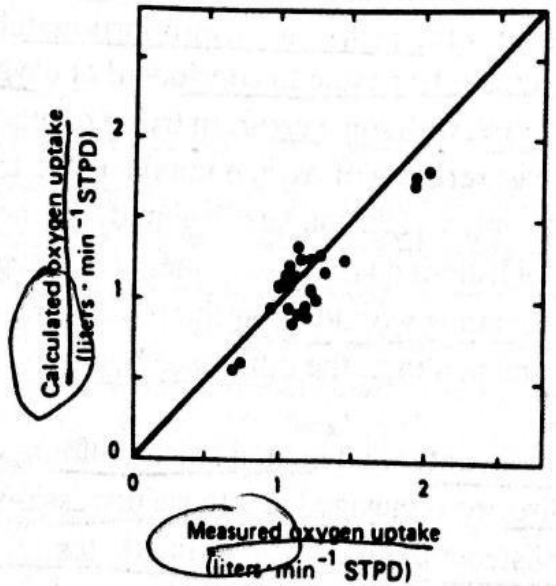


Figure 3.8 Relationship between O_2 consumption measured by the Douglas bag method and calculated from recorded heart rate. (From Rodahl et al. [14]).

Energy Cost of Work: Maximal Aerobic Power

- Maximal Aerobic Power (MAP)
  - Highest oxygen uptake an individual can attain during exercise (L/min)

- Good measure of physical fitness

- Young untrained male → 3.5 L/min

- Male rowers → 6 L/min

Energy Cost of Work: Maximal Aerobic Power

- Example
  - Task demand of oxygen uptake is 2 L/min
  - Person 1 has MAP of 4 L/min → 50% of the capacity
  - Person 2 has MAP of 2.5 L/min → 80% of the capacity
- Working at more than 30% to 40% of one's MAP in an 8-hour shift causes notable fatigue.

Work demands: Maximum Acceptable Limits

- For men, a maximum time-weighted average of:
  - 5 kcal/min for energy cost of work.
  - 100 beats/min for HR
  - 1.04 L/min for O_2 consumption
- For women, a maximum time-weighted average of:
  - 4 kcal/min for energy cost of work
  - 90 beats/min for HR
  - 0.83 L/min for O_2 consumption

TABLE 3.3 WORK CLASSIFICATION BY ENERGY COST

Work grade	Energy expenditure (kcal/min)	O_2 consumption (liters/min)
Severe	> 12.5	> 2.5
Very heavy	10.0–12.5	2–2.5
Heavy	7.5–10.0	1.5–2
Moderate	5.0–7.5	1.0–1.5
Light	2.5–5.0	0.5–1.0

Muscle Fatigue

- Fatigue: reduction in performance ability due to excessive physical/mental demands followed by inadequate recovery time.
- Many factors affect muscle fatigue development:
  - Strength
  - External forces
  - Posture
  - Contraction time
  - Recovery time
- Rohmert curve predicts that contractions below 15% MVC can be sustained indefinitely (10 min to > 1hr).

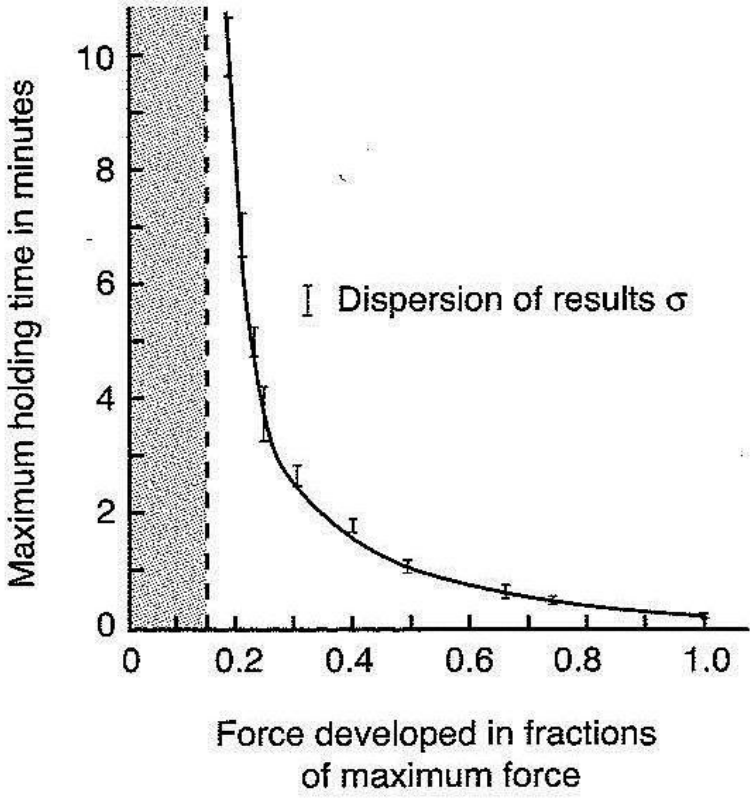
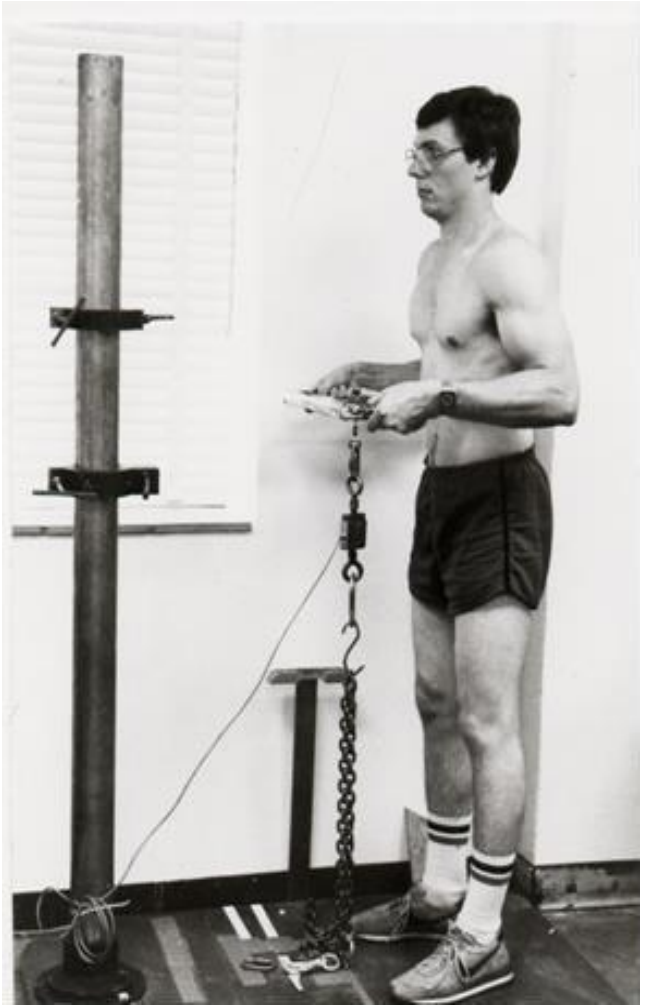


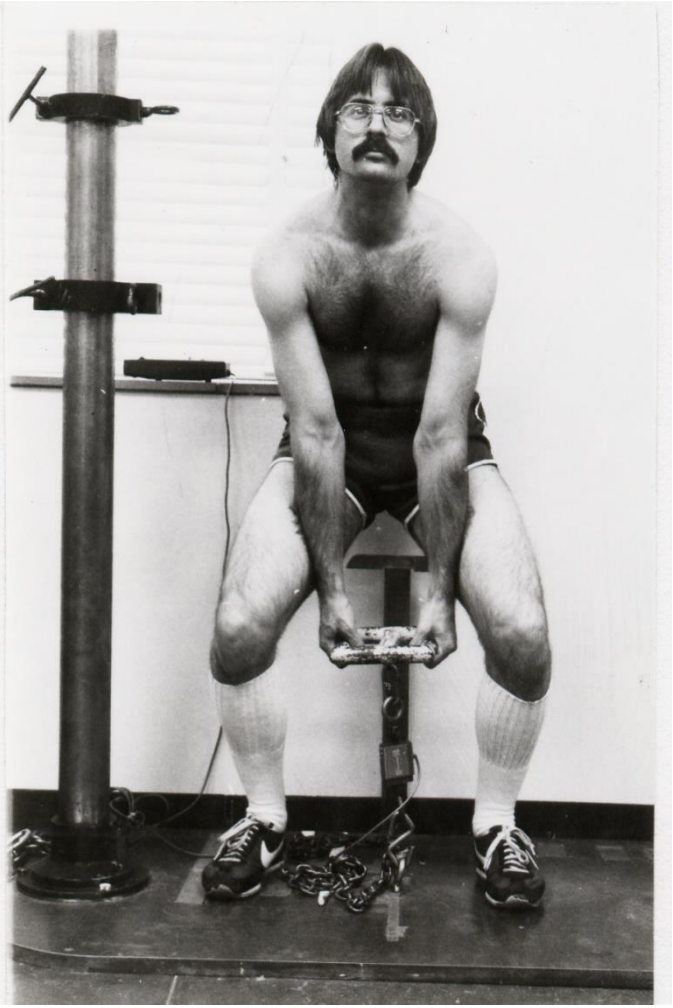
Figure 15.19 Maximal work time plotted versus force in isometric contractions. Average of results obtained in studies of different muscle groups. Fifty percent of maximal load can be maintained for just 1 min, whereas loads less than 15% of maximum can be maintained almost "indefinitely."

Reprinted, by permission, from W. Rohmert, 1968, "Die Beziehung Zwischen Kraft und Ausdauer bei Statischer Muskelarbeit. Schriftenreihe Arbeitsmedizin," *Arbeitshygiene* 22:118.

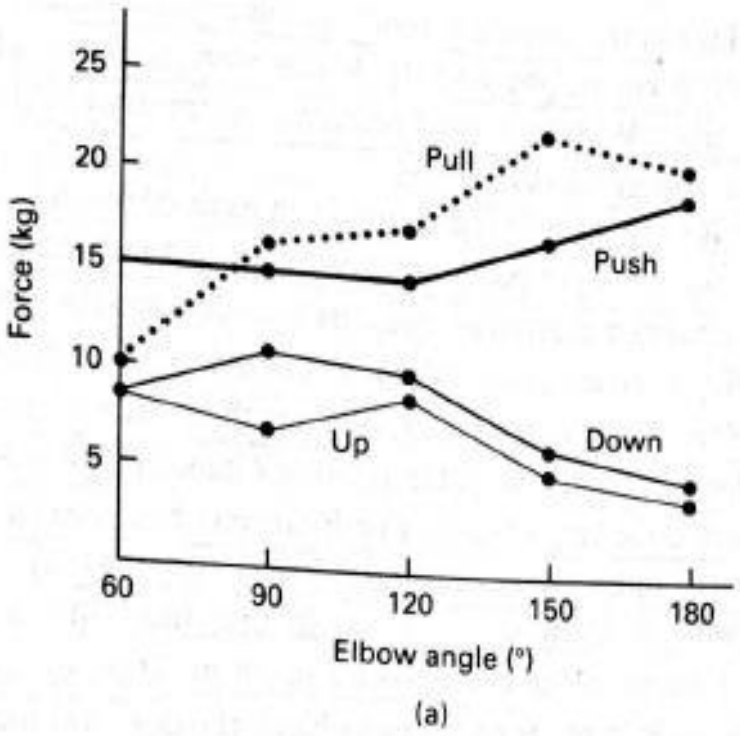
Strength

- Strength is the maximum force that one can exert voluntarily measured in kilograms (kg).
- Static strength is measured at static postures, while dynamic strength is measured during work.





Strength impacted by posture and force direction



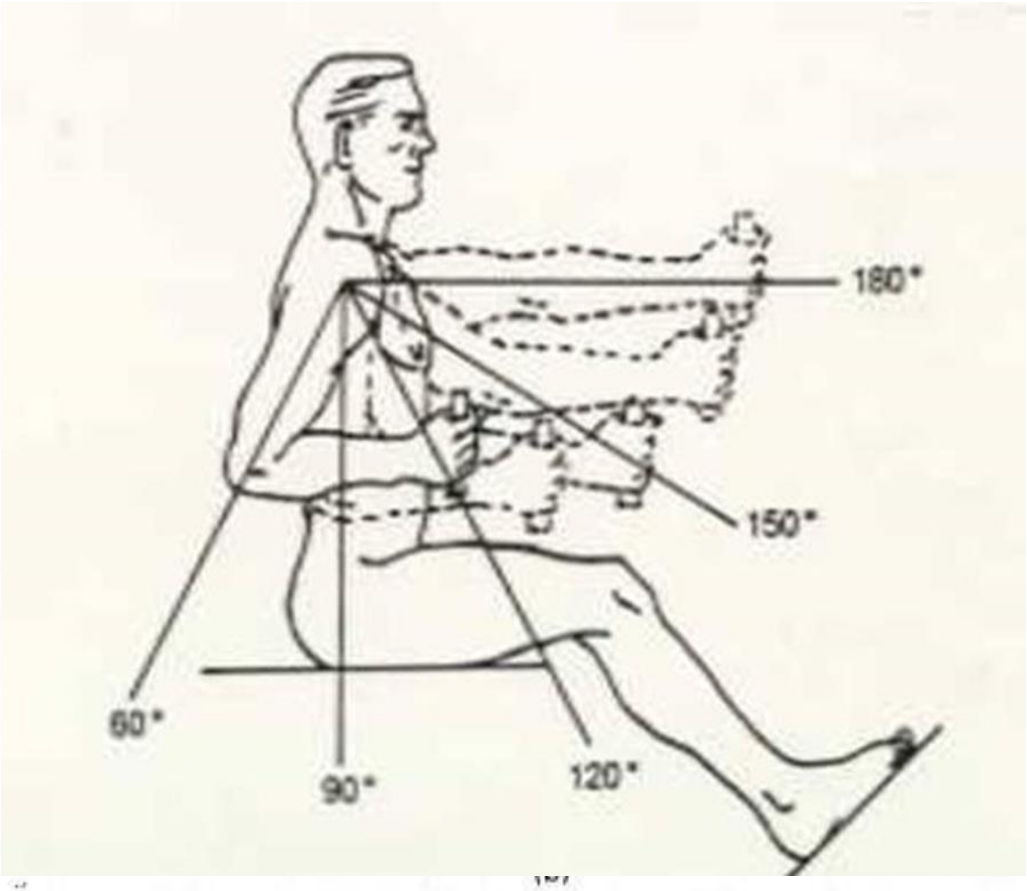


Figure 3.11 Strength in executing various arm movements in seated posture: (a) force exerted (5th percentile) by right hand; (b) elbow angles.

Strength

- It is estimated that for right-handed people, the strength of the left hand is approximately 10% below that of the right hand.
- Strength is at its peak by the late twenties and shows a continuous decline afterwards. At age 65, one has only 75% of the strength of a youth (~25-year-old).
- On average, women have two-thirds (66.67%) the strength of men.
- Exercise can increase strength and endurance by as much as 50%.
- Posture and duration are important factors.

Endurance

- Endurance is the ability to maintain activity over time.
- If an operator is expected to apply force continuously, the designer must consider endurance.
- Static effort can be maintained at only 20% of its peak over time, whereas 30% of peak forces can be maintained over extended periods in dynamic work.

Similar to Rohmert's Curve

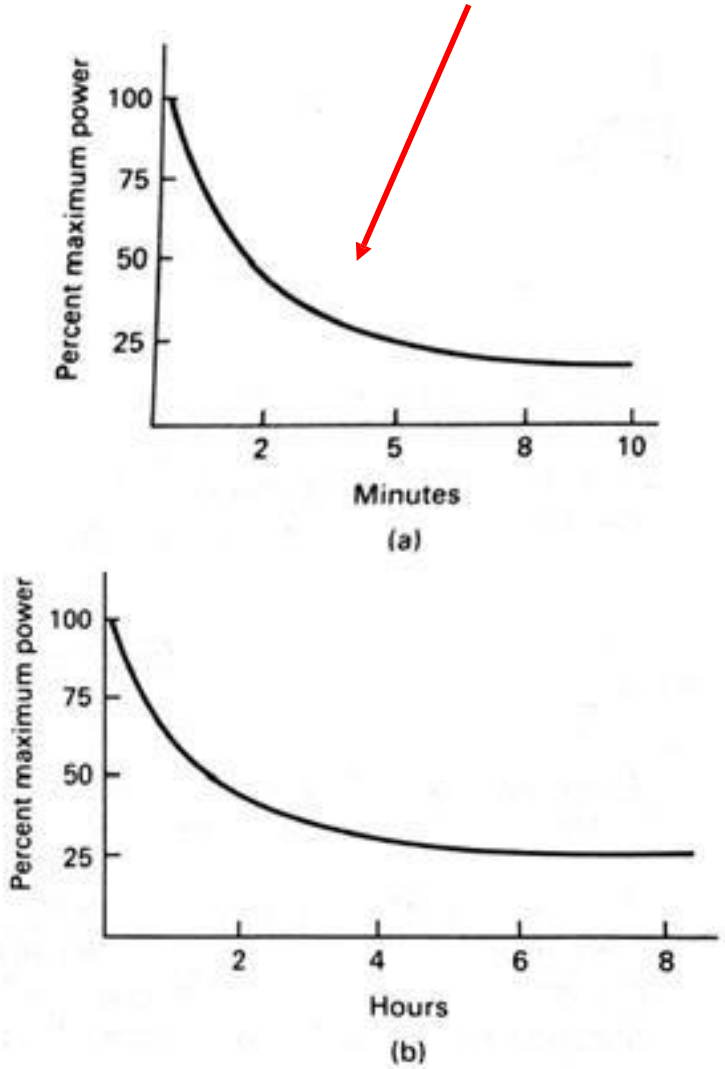


Figure 3.12 Endurance time as a function of maximum power: (a) static work; (b) dynamic work. (Reprinted with permission from *Human Factors*, Vol. 12(3), 1970. After Refs. [43] and [6]. © by The Human Factors Society, Inc. All rights reserved.)

Work-Rest Cycles

For K< S

$$R_T = 0$$

For S<=K< 2S

$$R_T = \frac{\left(\frac{K}{S} - 1\right) \times 100 + \frac{T(K-S)}{K-BM}}{2}$$

For K>= 2S

$$R_T = \frac{T(K-S)}{K-BM} \times 1.11$$

Where:

- K = the energy cost of work (kcal/min)
- S = the accepted standard (4 kcal/min for females,5 kcal/min for males)
- RT = the allowed rest time (min)
- T = the total expected duration of task (min)
- BM=basal metabolism (kcal/min)

Example 1

- Calculate the rest period for the shoveling task that requires 6.8 kcal/min performed over 1.5 hours. This task is performed by a male operator of basal metabolism 1.2 kcal/min.

For K>= 2S

$$R_T = \frac{T(K-S)}{K-BM} \times 1.11$$

For K< S

$$R_T = 0$$

For S<=K< 2S

$$R_T = \frac{\left(\frac{K}{S} - 1\right) \times 100 + \frac{T(K-S)}{K-BM}}{2}$$

Work-Rest Cycles

- Age affects the work-rest cycles

Age	Multiplier
20-29	1.0
30-39	1.04
40-49	1.1
50-59	1.2
60-65	1.25

- Age of the worker in Example 1 = 35 years

Example 2

- Calculate the rest period for a heavy assembly task that requires 9.2 kcal/min performed over 20 minutes by a 41 years old female operator with BM = 1.1 kcal/min.

For K>= 2S

$$R_T = \frac{T(K-S)}{K-BM} \times 1.11$$

For K< S

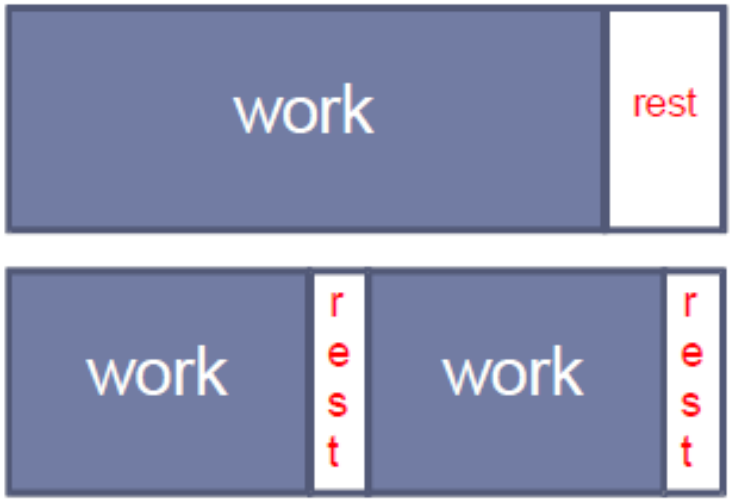
$$R_T = 0$$

For S<=K< 2S

$$R_T = \frac{\left(\frac{K}{S} - 1\right) \times 100 + \frac{T(K-S)}{K-BM}}{2}$$

Work-Rest Cycles

- An age allowance is necessary to consider after the RT calculation (as a multiplier).
- Research suggests that the most effective relief muscular fatigue is through small periods of rest taken several times during the duration of work rather than all taken at one time.



- It is customary in industry to allow for a 15- to 20-minute coffee break 2 hours after start in the morning and after lunch. These periods may be counted in the total rest duration.

UF

Herbert Wertheim
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LEADING THE CHARGE, CHARGING AHEAD